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Mobility versus job stability: Assessing tenure and productivity outcomes

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ABSTRACT

Based on the data from survey responses and publications of 1583 academic scientists in Spain, this paper examines the relationship between scientific performance and reward, considering tenure and permanent positions as key academic rewards in early phases of academic career and focusing especially on the mediating effect of mobile versus stable career paths. Although widely practiced, inbreeding has often been considered to be at odds with universalism and merit in science. Our findings indicate that inbred faculty does not get tenure with less scientific merits than PhDs from other institutions; we also find that non-mobile careers are a strong predictor of the timing of rewards in the form of early permanent positions. Our results question the assumption mainly based on US evidence that mobility enhances career. These findings must be interpreted in the context of organizational and institutional features of the Spanish academic system that promote the development of internal academic research job markets.

1. Introduction

This paper addresses the relationship between mobility of researchers, scientific performance and academic rewards in the form of tenure and permanent positions.[1] This relationship varies across countries. In US universities, academic achievement is strongly associated with mobility and there is some consensus on the rejection of departmental strategies to retain their new PhDs. In many other countries, systems of academic rewards are dissociated from mobility; recently China (Science, 2003), Japan (Nature, 1992, 2002), Korea (Nature, 1993; Science, 1998), and Spain (Nature, 1998; Science, 2006) among other countries, have attracted the attention of *Nature* and *Science* with editorials and reports criticizing inbreeding practices in their academic systems.

The debate on mobility and its desirability, however, is not new. The concerns about inbreeding in academia and its negative consequences emerged more than a century ago (Handschin, 1910), when it was a rather extensive recruitment practice, and a perceived problem among US universities. Charles W. Elliot, President of Harvard University between 1865 and 1909, stated that inbreeding (the recruitment for their faculty of people who had been granted their PhD by the same institution), though understandable, "was an undesirable practice for selection and recruitment at academic institutions" (Elliot, 2007). Later on, the general rejection of tenure track recruitment of PhDs granted by the university became an important institutional arrangement of the US academic system (Ben-David, 1984; Clark, 1995), despite recent research having analyzed that the reduction of inbreeding in top departments was the outcome of their increasing involvement in PhD exchange networks, a new form of social capital among elite universities (Burris, 2004).

Early empirical contributions in the US either tried to quantify the phenomenon (Hollingshead, 1938, 1940; McNeely, 1932) or showed the negative impact of the practice of hiring own graduates on university academic achievement (Eells and Cleveland, 1935a,b). However, beyond the normative rejection of the practice, in these early analyses there were very few why questions. While inbreeding has nearly vanished as a policy issue in US universities, in research the issue has been integrated into more general debates about the social structure of science and the role of mobility.

However, in many other countries[2] the debate around inbreeding is still relevant for policy and science studies; the extent to

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[1] We acknowledge that there are systems with and without tenure positions as such; however in this paper we use the term "tenure" to refer to permanent positions in general.

[2] We have found news and analyses regarding countries in Southern (Spain, Portugal, France, Italy, etc.) and Eastern Europe (Russia, etc.), Asia (China, Korea, Japan) and Latin America (Mexico, Argentina, Brazil, etc.) sharing this concern. Horta et al. (Horta et al., 2007) provide some examples.

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which these practices represent a contingent phase linked to the limited development of external academic markets or, alternatively, a different mode – not necessarily less productive or efficient – of organizing science is a key question. The structural features of academic systems (including incentives and governance mechanisms) should not be taken for granted or as universal; on the contrary, it should be acknowledged that diversity creates different models and dynamics (Clark, 1995; Musselin, 2004; Whitley, 2003). Empirical comparative research is required to establish whether "inbreeding" is a practice linked only to the early phases of development of academic systems or to low investment in R&D (Camacho, 2001), in cases in which institutions have not yet developed more universalistic approaches, job markets are not fully developed, salary levels are rigid, or in which the reputational systems of institutions have not consolidated.

Unfortunately, recent comparative analysis across countries has only attempted to measure levels of inbreeding and has been neither theoretically nor methodologically very sophisticated (Navarro and Rivero, 2001; Soler, 2001). Consideration of structural and institutional factors are largely missing in this type of studies, but inbreeding and aggregate low mobility levels are better understood in the context of the organizational dilemmas (Blau, 1994) between mobility and loyalty, between cosmopolitanism and job security, or between retention and turnover. These dilemmas are institutionally embedded. Shaped by structural factors, academic systems manage these dilemmas differently. In systems where universities cannot provide distinctive salaries or working conditions to their faculty members to reward achievement or in systems where there is not much reputational institutional differentiation, there is little incentive to mobility; we assume that academic systems develop responses constrained by their institutional arrangements, to compete in science and recruit researchers. There might be alternative models of academic careers, with significant differences regarding mobility, and a diversity of systems for managing human resources for research with different combinations of job security, salaries and mobility.

More comparative analyses are needed, as well as in depth country studies – that go beyond single discipline or research organization cases – to address the relationships among variables involved. The aim of the paper is to analyze the relationship between time to tenure, mobility and scientific production in the Spanish academic research system with emphasis on the impact of the phenomenon of inbreeding. These relationships are complex, and different institutional aspects should be considered to better understand and explain them. Institutional dynamics exist at organization-level, and implicit norms, expectations, and socialization within research groups and departments play a role in academic life. Additionally, research training systems and post-doctoral schemes are institutionally embedded (Gaughan and Robin, 2004) and must be taken into account in the analysis of careers, especially in early phases.

We first characterize the emerging patterns of academic careers in their early stages and look at the relationship of those patterns with differences in academic production. We specifically analyze the effect of mobility, inbreeding and other career variables on scientific output (publications) up to tenure. Secondly, we explain the probability of early tenure as a key reward in academic research and test the relevance of scholarly performance, mobility and other general variables.

The paper is organized as follows: in Section 2 we define the research questions and review relevant literature. In Section 3 we describe the data, variables and methods. In Section 4 we explain the institutional context of the public research system under analysis. In Section 5 we present the different statistical models and discuss the findings. We conclude summarizing the results, advancing some implications for research and policy and identifying further research.

2. Research questions and literature review

Our aim is to approach some of the dynamics of research careers in early stages by analyzing the relationship among tenure as a form of reward, mobility and scientific production. The relationships among these three elements are expected to be diverse in different institutional contexts. After describing the levels of inbreeding/mobility in the Spanish system, we want to address two main questions regarding this general objective:

- How does mobility affect scientific production up until tenure? We test the hypothesis that there are significant differences in performance between the inbred and the non-inbred faculty when they get their first permanent position.
- What explains early tenure as academic reward? We want to test the relative impact of past performance versus mobility and other career variables on the probability of getting a permanent position within 3 years after the PhD.

A large number of previous studies have addressed issues relevant to these questions. We first review some literature focused on the explanation of the two dependent variables: scientific production and academic rewards such as hiring and access to tenure. Secondly, we review some previous work on inbreeding and mobility of new PhDs and other relevant independent variables that should be taken into account in the explanation of scientific output and career advancement.

2.1. Productivity and early career progress

Empirical investigations that have approached the relationship between academic location, scientific productivity and rewards do not always coincide in their results and often have limited representativeness, especially when they focus on particular populations, cases or disciplines.

Publication productivity has been analyzed both as a dependent variable and as a causal factor in the explanation of other processes such as academic hiring or career advancement. When explaining scientific productivity as an outcome the literature has concentrated mainly on two different sets of factors: those related to environmental location and those referred to cumulative advantage processes.[3]

In the first set of literature, the institution's prestige appears as one of the strongest correlates of publication productivity. However the direction of causality is less clear and while they can operate in both directions, some longitudinal studies suggest that the effect of the organizational context on productivity is stronger over time than vice versa. In a comparison of different organizational contexts of scientific employment, including industry, Long and McGinnis (1981) found that within 3–6 years of getting a position in a specific context, scientists' level of productivity tends to conform to the characteristics of that *milieu*. Moreover, through the analysis of job changes, additional longitudinal research (Allison and Long, 1990) has shown that the effect of university affiliation on productivity is more important than the effect of productivity on university affiliation.

[3] There is a third set of literature related to individual-level characteristics (psychological traits, demographic characteristics and work habits) but these are strongly affected by the organizational and social context where they occur (Fox, 1983).

The second category of studies focuses on how feedback processes between productivity and prestige produce cumulative advantages and reinforcement effects. Allison and Stewart (1974) confirmed the hypothesis already advanced by Clemente (1973) that early publication in a career was the most powerful predictor of the publication record of an academic. In his studies, the rate of educational progress[4] and quality of the department appeared to have less impact upon publication productivity in contrast with earlier assumptions. But a negative relationship between the time taken to complete the PhD and subsequent professional success was also observed in several disciplines (Clemente, 1973; Hagstrom, 1971; Reskin, 1976).

The mainstream approaches in the study of the social structure of science consider the distribution of prestige among granting and hiring departments as a critical variable to explain hiring and early career advancement. Past performance, in the form of publication productivity, is often found as a competing explanation in these studies. For example, Crane (1965, 1970) analyzed the effect of the prestige of the PhD granting university and the scholar's publication record on the probability of getting hired in the top 20 departments for six disciplines. She found that, despite the system's normative commitment to universalistic criteria, those responsible for the recruitment of young faculty used the prestige of the doctoral department rather than past performance as a predictor of future performance (Crane, 1970). Additionally, she found high inbreeding rates among top departments. Cole and Cole (1973), Crane (1965), Hargens and Hagstrom (1967) found moderate and positive relationship between productivity and the prestige of the hiring department, mainly explained by the idea that the most prestigious departments tend to hire the most productive people; those findings appear to be consistent with normative expectations. Despite the inconclusiveness of these debates, there is consensus in the literature that the distribution of merit and rewards in the social structure of science is uneven and sometimes not coincident.

Long (1978) found that the effect of productivity on the allocation of positions was very weak. Long et al. (1979) showed that, in first appointments, departments usually break the rule of universalism and make decisions based on particularistic criteria, rather than on scientific production. They claim that the single most important element for recruitment and promotion from the point of view of universities and departments is not the candidate's "research record" but his or her "potential"; moreover Salancik and Pfeffer (1978) claimed that organizations make use of both universalistic and particularistic criteria, but are more keen on the latter in contexts of uncertainty when making a decision on substantive issues, and especially when allocating resources and hiring researchers.

2.2. Mobility of new PhDs, inbreeding and other explanatory variables

Mobility (or its absence) has been often analyzed as a factor explaining scientific productivity and academic advancements. It is worth remembering that the classic US studies of inbreeding[5] mostly found inbred scientists had poorer results and a slower academic progression.

Hargens and Farr (1973) published a seminal article in which they tested some of the main arguments developed by Berelson (1960) and McGee (1960); based on sample of 1165 academic scientists, they examined the relationships between academic inbreeding and scholarly performance, finding a "small but consistently negative relationship between being inbred and measures of scholarly productivity" (Hargens and Farr, 1973:1381).

From an organizational point of view, in his study of 115 American universities and colleges of all types, Blau (1994) called attention to the importance of loyalty and argued that the factor that exerts the strongest influence on a faculty member's loyalty to his academic institution is whether or not they have a tenure appointment. He explicitly introduced inbreeding as an independent variable and found that it also promoted loyalty. However inbreeding was found not to be positively correlated with an academic institution's reputation, qualification of faculty, research orientation or research productivity. Overall, the argument was that building a common academic tradition to increase loyalty to institutions might have negative side effects if inbreeding implies particularistic practices that affect judgments against merit criteria.

Recent contributions on inbreeding and their consequences are scarce; Wyer and Conrad (1984) in their analysis of more than 3050 questionnaires (345 of them answered by inbred faculty) from the Survey of the American Professoriate, found that "inbred and non-inbred faculty were not significantly different on standard measures of scholarly productivity" (Hargens and Farr, 1973:213) and that, after controlling for time allocation, inbred faculty were even more productive, but had lower salaries and less promotions.

Studies of faculty hiring and productivity in US Law Schools have continued to focus on the issue: Merrit and Reskin (1997) found that the most important factor to predict the probabilities of teaching in one of the top 16 Law Schools was inbreeding; they attributed this finding to the departments' risk-averse attitude and their perception that their own graduates were more attractive than graduates from other schools, had similar intellectual perspectives and were more loyal. Additionally Eisenberg and Wells (2000), studying 32 Law Schools and almost 700 entry-level faculties, found that the performance of inbred entry-level people was lower than that of those graduated from a different school.

In sum, answers to the question of whether inbred faculty produce fewer scholarly outputs than the non-inbred are often affirmative although findings are not so clear cut when controls on the quality of the hiring institutions or time allocation are introduced. What appears clearer is the fact that career advancement and access to tenure is slower among the inbred than among the mobile.

Other mobility forms could also affect the rate of career progress in different ways; international mobility, for instance, could be

[4] A set of variables refer to the speed at which individuals advance in the ladder of education; traditional elements like age at PhD or number of years elapsed between bachelor's degree and PhD were usually included in the analysis of productivity. The expectation is that the faster the progress the more productive the career.

[5] Some of the early studies on inbreeding should be mentioned: Caplow and McGee (Caplow and McGee, 1958) referred to two different departmental strategies regarding recruitment of new PhDs. On the one hand, inbreeding, the hiring of graduates in the same department in which they were trained, was commonly disapproved of but widely practiced; on the other hand, a twisted form of inbreeding called "outbreeding", consisting in hiring the graduates from other institutions for junior posts only (so that tenure appointments may be reserved for one's own graduates after they have been temporarily elsewhere) was characteristic of only a few of the top universities. Later on, Berelson (Berelson, 1960) addressed the issue in relation to the stratification forces determined by the reputation of the universities, arguing that the top universities, as a class, had to be inbred, as a statistical consequence of their dominant position as producers. However, Hargens (Hargens, 1969) using a subsample of Berelson's questionnaires, found that inbreeding rates did not vary greatly over prestige levels of universities. Studying the University of Texas, McGee (McGee, 1960) suggested that universities facing financial and geographical handicaps in their competition for faculty may appoint a larger number of their own graduates to junior faculty positions in order to liberate resources for hiring non-inbred more senior faculty in the hard competition of the national academic job markets. The arguments about some universities' difficulties to attract mobile faculty were stretched to sustain the existence of discrimination against inbred faculty, with lower salaries, longer periods for promotion, and poorer achievements. The methodological criticisms of early studies concentrated on the lack of use of multivariable techniques (Gold and Lieberson, 1961).

a mechanism of socialization in the international scientific community and could become associated with increased publication records (Aran and Ben-David, 1968; Reskin, 1977) but could have also negative effects (Melin, 2005); whereas keeping an homogeneous career pattern, as opposed to moving between academia and industry, has been found to yield a slightly positive effect in publication production (Dietz and Bozeman, 2005).

Among the individual attributes found to correlate with scholarly and academic market outcomes, sex is probably one of the most addressed in the literature on stratification in science (Fox, 1995; Fox and Stephan, 2001; Long and Fox, 1995; Stephan and Levin, 1992; Zuckerman, 1991). Most research finds that female researchers generally publish less than their male peers (Cole, 1979; Cole and Zuckerman, 1984; Reskin, 1978). However, some more recent research reports that differences in productivity by gender are diminishing (Xie and Shauman, 1998). In searching for causality of differential scholarly performance, some contributions argue that environments do not act neutrally on individuals and that their attainments are affected by the features of the organizational contexts in which they work (Fox, 1991; Fox and Colatrella, 2006; Fox and Mohapatra, 2007; Mählck, 2001). Regarding women's access to tenure, Fox and Colatrella (2006) use data collected in interviews with tenure and tenure track women faculty; although the majority of respondents reported that criteria for tenure were "somewhat" or "very clear", they also viewed the criteria as "unevenly applied" due to variations by field and research area and gender, among other variables. For Korean biochemists, Park (2007) finds that gender-neutral rules of academic performance can be applied in such a way as to maintain gender inequality in labor market output.

3. Data and methods

The strategy for building the data for this paper has acknowledged some of the methodological problems and limitations identified in previous research (Wyer and Conrad, 1984; Long and Fox, 1995; Chubin et al., 1981): too small N samples usually biased against the non-mobile, studies referred to a single scientific field or research organization, validity problems of self-reported productivity variables, single year studies and cross-sectional publication datasets instead of longitudinal data. In order to overcome these limitations we have built up a representative sample of the universe of reference, covering several scientific fields, we have gathered information about early trajectories and collected data on annual scientific production taken from generally accepted publication databases. Additionally we have used a diversity of techniques, combined data sources and matched individuals' records by gathering information from administrative sources, survey questionnaires, and Thomson-Reuters databases.

3.1. Data

The focus of our analysis was the timing of access to a permanent position and the relative scientific performance of these researchers in relation to some career variables. Data collection was guided accordingly. Our universe of reference were the faculty and researchers of all scientific fields who got their first permanent position between 1997 and 2001 at all Spanish public universities[6] or between 1997 and 2004 at the CSIC (Higher Council for Scientific Research).[7] University professors and CSIC scientists who get a permanent position are granted "civil servant" status; according to the "Register" of civil servants 7637 individuals got their first permanent position in all research domains at Spanish public universities, and 300 got permanence at the CSIC during the years of reference. In order to get valid results by scientific field (5) and size of the organization (5), a representative sample of 5306 individuals was selected from the database (Register of civil servants), including researchers from 38 universities and the CSIC, where similar rules for recruitment and promotion apply.[8]

Data for this study comes, on the one hand, from a national mail survey conducted in 2005, with a total of 2588 valid questionnaires (50% response rate) with a sample error of 1.58% and lower than 5% for the representative sub-samples. A structured self-administered questionnaire addressing research and professional trajectories, including more than 40 questions, was used to construct the different individual and career variables. For the present analysis, in order to guarantee the comparability of the measurement of scientific output (publications), we dropped individuals from social sciences and humanities.[9] The final size of our dataset for this study was 1583 individuals (32.5% of them women, representing the same percentage as in the universe of reference), the majority of whom are in their mid careers (mostly in their early forties, with an average age in 2005 of 42 years old, a median and mode of 41, and with an average of 6.7 years after tenure) and covering three broad different scientific fields: Biological and Medical Sciences, Exact and Natural Sciences and Engineering and Technological Sciences.

On the other hand, in order to complement the information obtained from the questionnaires and considering scientific publications in peer review journals are generally accepted as one of the most important elements for career (Fox, 1983), we constructed a database of the individuals' pre-tenure publications record in journals between 1990 and 2004 included in the Science Citation Index Expanded – SCI – from Thomson-Reuters, by matching the names of the individuals in our survey.[10] We have included all types of publications that match the way authors sign their papers. We did not use fractional counting: a single publication with 20 co-authors, 10 of them in our sample, was counted 10 times and attributed once to each researcher. Although this data source allows for the possibility of analyzing citations, for the purposes of studying early careers, we have not included measures of impact associated to citations. The reason is twofold: firstly, to avoid the problems of making comparisons across disciplines that have diverse citation models (Cozzens, 1985; Leydesdorff, 1998), and secondly, because we accept the findings of Long et al. (1993) regarding the quantity of publications being more important than quality in predicting access to the first tenure position. Since our focus in this paper is on access to tenure we do not include post-tenure publications here.

3.2. Methods and variables in the analysis

Descriptive statistical analysis was conducted in order to identify the basic indicators of research production and the distribution

[6] In Spain, public universities represent almost 92% of the students' market, and more than 95% of research activity at higher education institutions.

[7] CSIC is the largest Spanish public research performing institution, similar to the CNRS in France or the Max Planck Society (MPG) in Germany.

[8] We do not analyze here individuals who did not succeed in getting the permanence because the focus is not tenure versus non-tenure as a labor market outcome. A comparison between successful and failing candidates was not feasible methodologically; however such a comparison would be useful to control for the possibility that the selection was biased against very productive non-inbreed scholars.

[9] Only a limited number of researchers in those fields (less than 15% in our original sample) had any of their publications included in these databases, confirming the different publication patterns of the majority of academics in Social Sciences and Humanities and the difficulties of analyzing scientific performance in these fields based only on international papers (Nederhof et al., 1989).

[10] Using individual CVs as a source of data for the analysis of career trajectories is an emerging field of methodological development (Dietz et al., 2000); however we only used certain key data in the questionnaires to match the data with the publications database.

Table 1
Description of the variables.

Variable	Description	Observations
Age in 2005	Quantitative variable representing the age of individuals at the time of survey	
Year of PhD	Quantitative variable representing the year in which the individual was granted the PhD degree	In some cases the variable has been transformed into a quantitative variable ranging from 1 to 28 (the maximum scope of our sample), meaning 1973 (1) and 2000 (28)
Time to PhD degree	Quantitative variable calculated to represent the number of years elapsed between getting the bachelor's degree and getting the PhD	It is used as a proxy of the rate of educational progress
Sex	Male or female	
Mobility to a different center right after the PhD	Dummy variable measuring whether or not the individual's first employment after the PhD was in a different university/center than the one that granted him/her the PhD or in which he/she was working at the time of getting the degree	
Mobility outside academia in the first job after PhD	Dummy variable measuring whether or not the respondent got his/her first employment after the PhD in a non-academic or non-research organization	
Post-doctoral international mobility	Dummy variable that measures if the respondent reports academic stays abroad for at least 6 months during the period after the PhD and before obtaining tenure	
Publications until tenure	Measure of scientific production over time of individuals, includes publications before and after the PhD degree; it is a quantitative variable representing the number of times the name of an individual appears as author in papers, regardless of the number of co-authors	In some of the regression models this variable was transformed into its natural logarithm to cope with the skewness of the distribution
Annual publications until tenure	Measure of individual scientific productivity per year; it is a quantitative variable representing the number of times the name of an individual appears as author in papers, regardless of the number of co-authors, divided by the number of years elapsed since his/her bachelor's degree	In the Logit regression model this variable was transformed into its natural logarithm to cope with the skewness of the distribution
Early publication	Dummy variable constructed with the information from the record of scientific publications for each individual, and measures whether or not the individual has authored at least 1 publication on or before the year he/she was awarded the PhD	
Tenure in a center different from the center that awarded the PhD (non-inbred status)	Dummy variable measuring whether or not the respondent got tenure in a different university/center from the center that awarded him/her the PhD degree. It is the variable we used to measure inbreeding	We apply, as Berelson (1960), the concept of "inbred" to the people that got tenure in the same place they were awarded the PhD; therefore, it also includes the so called "silver corded" scholars, who, in Spain, are likely to be individuals who have experienced temporary mobility sponsored by the granting departments
Early tenure	Measure of the access to lifetime employment early in the career; it is a dummy variable that measures whether or not the individual got tenure within 3 years after obtaining the PhD degree	
Ranking position of the center of tenure	Measure of the relative research quality of the institution (not the department) that has hired the individual	Since there are no standard reputation rankings for Spanish universities or research institutions, we have used data on research competitiveness; the indicator represents the relative position of each organization with regard to the ratio of success in applications to competitive funding projects over a period of 6 years; see García and Sanz-Menéndez (2005) for details. It is a scale variable ranking from 1 to 39, which allocates a position to each Spanish university or CSIC in the sample of hiring institutions, based on the aggregate success rate in competitively funded projects ranking from 1 (the highest) to 39 (the lowest)
Research field	Is a classificatory variable of the researcher's main domain of activity. It follows the criteria set up by the OECD Field of Science Classification (FOS)	Our aggregation level within this classification is the highest (field), and the 3 possible values are: Biology and Biomedical Sciences (used as a reference in the regressions); Exact and Natural Sciences; and Engineering and Technological Sciences

of several variables related to institution of tenure, inbreeding status, relative degree of mobility, employment trajectory, research field, age and sex. More explanatory tests are presented in relation to the factors underlying publication production and early tenure, looking especially to the role of different forms of mobility.

We have used multivariate statistical techniques of analysis. First, we ran a multiple regression replicating the Hargens and Farr (1973) test to explain the effect of being inbred (getting the tenure in the same department that granted the PhD) and other career trajectory and individual variables in the number of publications up to the tenure. We tried two different transformations to cope with the skewness of the distribution of the number of publications: the square root and the natural logarithm; we opted for the latter. It must be acknowledged, however, that any of these transformations imply some diminishing returns to productivity. To cope with the skewness problem we also ran a negative binomial regression[11] to explain productivity. Secondly, we used a Logit model to analyze the factors associated with early tenure because logistic regression is well suited for the study of categorical or dichotomous outcome variables.

All variables included in our different models are described in Table 1. For the basic descriptive statistics, see Table 2.

Table 2
Descriptive statistics of the variables.

Quantitative variables	Mean	Median	S.D.
Age in 2005	42.01	41.0	4.8323
Year of PhD	1993	1994	3.5474
Time to PhD degree	6.47	6.0	3.2264
Publications until tenure	27.03	11.0	54.1687
Annual publications until tenure	2.40	0.9	5.3633
Ranking position of the center of tenure	18.13	17.0	12.0289

Categorical variables	Mean
Sex (reference: women)	0.33
Mobility to a different center after the PhD (reference: yes)	0.31
Mobility outside academia in the first job after PhD (reference: yes)	0.03
Post-doctoral international mobility (reference: yes)	0.58
Early publication (reference: yes)	0.59
Tenure in a center different than the one that granted the PhD (reference: yes, non-inbred)	0.39
Early tenure (reference: yes)	0.28
Research field	
a: Biological and Biomedical Science	0.31
b. Exact and Natural Sciences	0.37
c. Engineering and Technological Sciences	0.33

$N = 1583$.

4. Institutional and structural features of the Spanish academic and research context

In this section we frame our analysis within the specific structural forces of the Spanish academic and research system and the constraints and incentives that organizations and individuals face as regards hiring and granting of permanent positions. Additionally, the degree of development and the size of the market of researchers and new PhDs in the different scientific fields and its fragmentation in areas are important elements to consider.

We need to understand the differences regarding how positions exist or are created in different academic systems and how the jobs might be allocated. Long et al. (1979) argued that positions in the US academic jobs market "are not allocated by some master decision makers, but by a market mechanism in which departments compete for the most desirable candidates; candidates decide which jobs to apply for, departments choose candidates for the vacancies, and candidates choose among the offers they receive" (Long et al., 1979: 817). The candidates themselves play a large role in the final outcome.

In the systems where tenure positions exist, open jobs are designated eligible for tenure, or "tenure track", during the hiring process. Typically a professor hired in a tenure track position will work on a probationary basis for approximately 6 years after which an evaluation is made and a tenured position is eventually granted or denied. The pre-tenure period is meant to provide the candidate with opportunities to reveal competencies (Musselin, 2005).

In other countries, the procedures for hiring, selecting and providing permanence to researchers differ from the model described.[12] Human resource management is a highly centralized function in Spanish and other European academic institutions as well as in some Asian ones, but it not is the case in the US. In the former countries, permanent positions are usually created *ex novo* by the university authorities. In general, the departments are not endowed with the resources to manage permanent appointment decisions or create new permanent positions independently. If a permanent faculty member leaves the institution, the position is often completely lost for the department and new rounds of negotiations with authorities start from scratch to try to get a new one.[13] This context of authority distribution creates some pressure for the department to support and reward "loyal candidates", who will not leave or go away; in doing so, departments minimize the risk of losing positions that are costly to get.

In many academic systems it is not accurate to talk about "tenure track positions" and the traditional entry point to the academic labor market is either a lecturing position or a PhD student research grant. Traditionally, in Spain, as in France (Gaughan and Robin, 2004), research training policy was more likely to support the individual graduate student directly through a national or regional grant instead of indirectly through a research grant given to principal researchers or institutions. There exist implicit contracts between the PhD candidate and the PhD supervisor, mutual expectations about collaboration and recruitment that occur in the research training period.[14] Thus, the entry in the labor market often takes place very early in the research career, and this period is more

[11] Negative binomial regression models are considered appropriate for modeling events that, as is the case of the distribution of our dependent variable, depart markedly from a symmetric normal distribution, are characterized by overdispersion, and in which the sample variance exceeds the sample mean, even if there are some caveats (Allison and Waterman, 2002). Poisson regression models are also often used to model skewed distributions. We chose negative binomial instead of Poisson because the latter model places a number of constraints on data to which it is applied, which were not met by our data. First and most importantly, the Poisson regression model assumes that events (those measured by the dependent variable) occur independently over time and that there is no kind of dynamic dependence between the occurrence of successive events. It would be difficult to claim that the events we are observing in this case, that is, publishing articles, occur independently over time. This would be not only against commonly shared theoretical claims about cumulative advantages in science, but also against most of the empirical evidence on the issue. Additionally, the Poisson distribution assumes that the mean and the variance are identical, which was not the case of our publication variable. The inappropriate imposition of that restriction on the data leads to the underestimation of the standard errors and spuriously inflates the tests of significance (Land et al., 1996). We tentatively ran the model and confirmed that, in effect, variables that were not significant either in negative binomial or linear models appeared as significant in the Poisson. In any case, we thank Hugo Horta for suggesting we try this model.

[12] Perhaps one of the key differences is that those procedures are governed by universal compulsory legal rules with almost no freedom for local procedures.

[13] In some US universities this practice is becoming more common as result of the difficult economic situation.

[14] An indicator of the strength of the social ties developed between PhDs and their supervisors in Spain is the fact that 88.3% of the respondents in our survey reported that they had published with their supervisors in the predoctoral period, and 79.7% reported they had collaborated with them in research projects after getting the PhD.

strongly associated with learning rather than with demonstrating performance. In our population, the average age of entry in the institution that granted the permanent position was 28 years old, while the average age at PhD was 31 and the average age at tenure was 36.

In Spain, permanent positions must be published in an open call to which anyone with a PhD degree (the only formal requirement) can apply. Applicants may come from the department or be in another university or research center. In our period of reference, applicants were evaluated by a committee comprising five members, usually two from the department to which the new position had been allocated and three selected randomly from all permanent faculty members in Spanish universities in the same disciplinary area[15]; the Chair of the selection committee was often one of the two members of the department. This selection procedure was established in 1983 in the University Reform Act (Sánchez-Ferrer, 1997); at the end of 2001 a new Universities Act significantly changed the procedure for selection and granting permanent positions,[16] which moved from a significantly decentralized scheme dominated by the hiring department into a national accreditation and selection scheme with single committees for all candidates and covering positions in all Spanish public universities (Cruz-Castro and Sanz-Menéndez, 2007). Rules and incentives operate in the selection process and influence the degree of closure or openness of the academic job structure, favoring the construction of internal labor markets in different degrees.[17] The number of potential competitors (in and out of the department) may vary. It is often the case that there are queues, or an informal order of priority in the advancement of career (in some cases based on merit and in others on the time spent at the department). This system implies high transaction costs for those not in the department, because academic cultures tend to value loyalty highly and tend to reward it accordingly. It often appears risky for outside applicants to invest time and effort in preparing the selection process without clear chances (or without the support from the department).[18]

An additional key element to consider is that there are no salary negotiations in academic recruitment. In Spain, as in many other European countries, permanent university professors and CSIC researchers are civil servants (Mora, 2001) and in Spain, their salaries are set up on the basis of bureaucratic rules: equal categories earn the same amount and some minor differentiation come from research productivity complements (Sanz-Menéndez, 1995). This is important because it limits the ability of organizations or departments to bargain with the job candidates in the recruitment and selection process. What is at stake is permanence and promotion through scales; those are the main assets that departments, universities and research centers can use for the management of their human resources.[19] Permanent positions are the main rewards. It is, therefore, a strongly regulated system, with very few market mechanisms. In fact, the model fits in the trade off between tenure probabilities and salaries described by Ehrenberg et al. (1998) in relation to US Economics Departments that offer lower probabilities of tenure, but are willing to pay higher salaries to post-docs or assistant professors. In our case, the trade off would be the opposite.

A final constraint to the existence of an open academic and research job market is the way universities are financed. The regional governments finance public universities in their territories. The distribution of resources is made according to the number of students and the type of degree (technical or other) and the teaching loads (González Lopez, 2006). Universities do not need to compete for the best professors to keep going or to advance financially, because the contribution of research overheads to the funding of universities and research centers is small. These dynamics are embedded in a context in which evaluation of performance at institutional level is marginal (Cruz-Castro and Sanz-Menéndez, 2007).

The overall low degree of mobility in the Spanish system and in other European ones (Ackers, 2008) should be understood within the institutional context described and was identified as a policy issue long ago (Brickman, 1977).

The empirical results coming from our survey not only provide evidence of a significant level of inbreeding, but they also show signs of the existence of internal labor market dynamics starting in the predoctoral period. There is a very limited degree of early post-doctoral mobility across universities. 69.0% of the PhDs did not change their employer after getting their PhD.[20] A further indicator is that 45.8% of the surveyed population stayed at the same university for their whole career trajectory (bachelor's degree, PhD and permanent position). Six out of ten individuals (60.8%) got a permanent position in a department located in the same university that awarded their PhD diploma.

5. Mobility, past performance and organizational rewards: analysis and findings

To assess the association between inbreeding and performance we first ran a multiple regression model in two phases (see Table 3). The first model regresses the natural logarithm of the number of publications until tenure and introduces the scientists' inbreeding status, a measure of university quality, and the year of the doctorate as independent variables.[21] We wanted to test whether there are significant differences in the publication production between those who got their PhD degree from the same institution where they got tenure and those who did not. It is well known that the quantity and quality of a researcher's publications are moderate but positively associated with the prestige of the hiring department. We do not have a measure of prestige for our research institutions, but we do have a measure of the relative research competitiveness of our sample of organizations, and we ranked them accordingly. The year of doctorate was included to account for the fact that scientists who have launched their careers as PhDs later might have a lower publi-

[15] There are two hundred knowledge areas from which members for the hiring committees are selected. This disciplinary fragmentation makes the average size of the area rather small, and contributes to reducing competition, and to fostering the existence of rather closed networks and dynamics that tend to favor departmental preferences regarding candidates.

[16] In fact, that circumstance was a criterion for the selection of our universe (those getting a permanent position at the universities between 1997 and 2001), in order to ensure that all selected individuals were subjected to the same rules.

[17] The concept of internal labor market (ILM) was first developed by Doeringer and Piore (Doeringer and Piore, 1971) to describe the existence of some firms in which pricing and allocation of labor is governed by a set of administrative rules and procedures. The positions within the ILM are filled by the promotion or transfer of workers who are already inside; the ILM would thus be protected from the competition of external labor markets. Entry in the organization is at the low levels of the professional ladder, on-the-job training is very relevant and promotion is strongly linked to seniority.

[18] An additional indicator is that in 58.4% of the competitions there was only one candidate. The percentage is higher for universities (66.3%) but very low at the CSIC (only less than 9.3% had no competitors for the permanent position).

[19] The Spanish system is not an "up or out" one; candidates can apply for a permanent position more than once. Most of the researchers within the academic system get the permanence at some stage of their career (unless they abandon).

[20] Moreover, 42.3% of the individuals who got the permanence in the period under study did not report any kind of international post-doctoral experience. However, for those who moved abroad temporarily, access to permanence back in Spain was not direct; in almost all cases, they came back to a non-permanent position first and got permanence afterwards.

[21] Hargens and Farr (Hargens and Farr, 1973) used similar variables in their classical contribution about inbreeding.

Table 3
Publications until tenure (OLS equation).

Variables	Model 1			Model 2		
	Unstandardized coefficients			Unstandardized coefficients		
	B	S.E.	Exp(*B*)	*B*	S.E.	Exp(*B*)
Tenure in a center different than the one that granted the PhD (reference: non-inbred)	0.142*	0.079	1.153	0.040	0.060	
Year of PhD	−0.046***	0.011	0.955	−0.121***	0.009	0.886
Ranking position of the center of tenure	−0.011***	0.003	0.989	−0.007**	0.002	0.993
Early publication (before PhD) (reference: yes)				1.967***	0.063	7.149
Post-doctoral international mobility. (reference: yes)				0.262***	0.061	1.300
Sex (reference: women)				0.062	0.062	
Time to PhD degree				−0.009	0.009	
Research field (categorical). Biology and Biomedical Sciences (reference category)						
Exact and Natural Sciences				0.115*	0.061	1.122
Engineering and Technological Sciences				−0.040	0.082	
Constant	3.447	0.245		3.666	0.196	
R	0.158			0.671		
R^2	0.025			0.451		
R^2 corrected	0.023			0.449		
Valid cases:	1450			1450		

Note: Unstandardized coefficients – *B* – are referred to the dependent variable (natural log of publications until tenure), but Exp(*B*) are referred directly to the number of publications until tenure, but calculated only for significant variables.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.001$.

cation record just because they have fewer years to publish before tenure. Overall the model has a very poor explanatory power (only 2.3% of the variance of the dependent variable), and the coefficients are very modest. In this model the inbreeding status of the scientist, being only significant at 0.10 level, is not a strong predictor of publication record until tenure; the impact is small considering that the non-inbred are expected to produce 15.3% more papers until tenure than the inbred scientists.

The second regression model adds some other theoretically relevant independent variables: early publication, time to PhD degree, international post-doctoral mobility, sex, and scientific field. Overall, model 2 is significant and explains about 45% of the variation of publication productivity. However, in this model, inbreeding status, time to degree, gender and the technological sciences field are not significant variables. The first important thing to note is that once the rest of the variables are specified in the equation, the inbreeding status is not significant any longer. Therefore, the results do not provide conclusive empirical evidence in support of the claim that those who left their doctorate university to get tenure at another institution are more or less productive than those who did not. The significant variables are of a different nature. Having published at least one paper before being granted a PhD showed the strongest significant independent effect on the overall publication record up until tenure; net of other variables, those who published before being awarded their PhD produced almost seven times more papers than those who did not. This result is coherent with the claim regarding the accumulation of advantages along the career.

The results suggest that those with earlier doctorates had published more at the time of getting a permanent position. Also as expected, being tenured in an organization high in the ranking of success in competitive research funding projects is positively related to publication productivity of scholars up until the tenure. One measure of mobility is the variable of post-doctoral research stays abroad. International experience means access to wider and more open scientific networks which might facilitate production and publication; this mobility variable appears modestly yet significantly and positively linked to publication productivity.

The lack of significance of some theoretically relevant variables was unexpected. As an alternative method to cope with the statistical problems emerging from the skewness in a different way we ran a negative binomial regression replicating model 2, but with the untransformed dependent variable of total number of publications up until tenure. The results are shown in Table 4. The model yields similar results but in this case the sex, scientific field (all fields) and time to degree variables are significant. Longer PhD completion times are negatively associated to the total number of publications. Women have also published less up until tenure, but the effect of this variable is just modest and significant only at 0.10 level. Scientific areas have different publication habits; we found that academics in technological sciences show a larger number of publications before tenure, followed by those in exact sciences, who in turn are followed by biologists and medical scientists – the reference group – as the least productive until tenure. The inbreeding variable is not significant either in the negative binomial model confirming our former claim about the lack of differences between both groups. Neither the idea that inbred faculty are less productive than their non-inbred colleagues, nor the idea that they are more productive, find much support in the models.

After reaching a tentative conclusion about our first research question, we now turn to the issue of academic rewards, a crucial one in the institutional context under analysis: early tenure. A permanent position is a highly public reward in science, and as far as it is a form of recognition, it should be governed by the norm of universalism. Within universities and research centers, one of most highly valued activity is contributing to the body of certified knowledge through publications. One could expect scholarly performance in the form of publications to positively affect career advancement. But we might wonder about the impact of inbreeding and mobility within organizational contexts that favor socially constructed internal job market dynamics. How important is scientific performance in explaining early tenure?

In order to analyze the relationship between performance, mobility and organizational rewards, we looked at the probability of getting tenure in the associate professor position within 3 years or less from completing the doctorate.[22] Besides the gen-

[22] This might seem very early but the median and the mean of the distribution of the variable measuring the number of years elapsed between the PhD and the tenure is five.

Table 4
Publications until tenure (negative binomial regression).

Variable	Unstandardized coefficients		
	Beta	S.E.	Exp(Beta)
Tenure in a center different than the one that granted the PhD (reference: non-inbred)	−0.066	0.073	0.936
Year of PhD	−0.105***	0.013	0.901
Ranking position of the center of tenure	−0.007**	0.003	0.993
Early publication (before PhD) (reference: yes)	2.388***	0.087	10.894
Post-doctoral international mobility (reference: yes)	0.175**	0.077	1.191
Sex (reference: women)	−0.121*	0.075	0.886
Time to PhD degree	−0.037***	0.268	0.963
Research field. Biological and Biomedical Sciences (reference)			
Exact and Natural Sciences	0.164*	0.086	1.178
Engineering and Technological Sciences	0.226**	0.101	1.254
Constant	1.449		
Chi-square	630.04		
Pseudo R^2	0.053		
Valid cases:	1450		

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.001$.

Table 5
Tenure within 3 years after PhD (logistic regression).

Variables	Beta	S.E.	Exp(Beta)
Annual publications until tenure (ln)	−0.287**	0.103	0.751
Tenure in a center different than the one that granted the PhD (reference: non-inbred)	0.358*	0.180	1.430
First post-doctoral job outside academia (reference: yes)	2.309**	0.838	10.069
Change of center after the PhD (reference: yes)	1.245***	0.232	3.474
Post-doctoral international mobility (reference: yes)	0.635***	0.171	1.888
Sex (reference: women)	0.987***	0.184	2.684
Age in 2005	−0.451***	0.036	0.637
Time to PhD degree	0.510***	0.045	1.665
Research field. Biological and Biomedical Sciences (reference)			
Exact and Natural Sciences	0.799***	0.244	2.223
Engineering and technology Sciences	1.863***	0.239	6.891
Constant	8.801	1.452	6640.066
N valid = 1496			
R^2 Nagelkerke = 0.565 and 85.1% correct classification			

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.001$.

eral variables of sex, research field, time to degree and age, we introduced in the model one variable measuring past performance (annual publications up until the tenure, transformed into its natural logarithm) and a group of variables related to relative mobility, including inbreeding status and other variables measuring different types of mobility such as: whether the first employment after the PhD was outside academia, whether the individual changed center in the first year following the doctorate, and whether he or she spent some research time (minimum 6 months) abroad after the doctorate and before getting tenure. Logistic regression results are shown in Table 5. The model is significant, has a fairly good adjustment and classificatory power.

The first thing to note is that more productive individuals do not seem to be rewarded with early tenure.[23] In fact, the coefficient for annual publications until tenure is negative.[24]

The explanation of early access to a permanent position should be found in other variables. Turning to the variables related to first job placement and mobility, it stands out that the organizational context of the entry job after the PhD has a significant effect on advancement in academia at later stages.[25] Those who had their first employment after the PhD outside academia are in disadvantage for early tenure in comparison with those whose first position as PhD holders was at a university or a public research center[26] (the odds of early tenure for this latter group are in the order of nine times higher, but the distribution shows high levels of instability).

In academic organizational structures characterized by the existence of internal job market dynamics, other mobility variables would be expected to be negatively related to the dependent variable. This is confirmed by the model: early tenure does have an association to all forms of mobility, and it is negative, even when controlling for annual productivity. It is clear that inbred and non-mobile faculty are in relative advantage to get early tenure with

[23] In previous literature, the lack of relationship between productivity and job placement has usually been related to entry jobs (non-tenured) and not to advancement in rank (access to tenure or promotion to from associate to full professor).

[24] We used this productivity measure (number of articles per year) in order to control for some likely time effect. We also tried the model with the total number of publications. This alternative measure also showed a negative coefficient in a very similar model. Since the variables ‘total number of articles’ and ‘number of articles per year’ had a strong correlation coefficient (0.9) we decide to introduce only the last one.

[25] There is not a significant relationship between this variable and those measuring productivity, therefore effects can be regarded as independent.

[26] The negative effect of non-homogeny in careers was also found by (Dietz and Bozeman, 2005) in their analysis of the effects of mobility between academia and industry on publication productivity.

respect to non-inbred and mobile.[27] The odds of getting tenure in 3 years (compared with not having early tenure) are increased by a factor of 1.430 by being inbred rather than non-inbred; however we should mention that the variable is only barely significant ($p = 0.046$).

The same applies to those who did not have international research stays abroad, and did not move to a different center or university in the year following completion of their PhD. The odds of getting early tenure are increased by a factor of 1.888 by not having post-doctoral international mobility rather than having it. Additionally, the odds of getting early tenure are increased by a factor of 3.474 by not changing the center after the PhD rather than changing.[28]

Looking at the first five independent variables together we observe not only that past performance does not explain early tenure, but also that few incentives are given to international or national mobility which merely delays access to tenure jobs.[29]

As regards the more general variables, we can observe that the odds of getting tenure early in the career are more than double for male researchers than for their women counterparts.[30]

Older individuals are less likely to have obtained their first tenure position early after the PhD. Interestingly, in contrast with most studies, longer periods from bachelor's degree to PhD do not have a negative impact on the career. Quite the opposite, the coefficient for this variable is positive, reflecting subtle dynamics at the level of the departments; once the scholar who has often been assistant professor for many years is in possession of the formal qualification required for the post, these dynamics speed up the process of granting them with a permanent position. The odds of getting tenure in 3 years compared with not getting early tenure are increased by a factor of 1.665 by every additional year taken to get the PhD. This finding suggests the existence of internal labor markets dynamics and a strong mechanism of rewarding loyalty.

We also found significant differences by scientific field. Taking the biologists and medical researchers as the reference group, the odds of getting early tenure for exact and natural scientists are more than double, in fact they are increased by a factor of 2.223; for engineers the advantage is even higher. This might be reflecting differentials in the labor market situations for different areas and the increases in the supply of life science doctorates over the nineties that even a growing demand could not cope with. Also it might be the case that engineering departments have extreme retention practices taking into account that their graduates have many more alternative career opportunities outside academia.[31]

Both the subgroup of women and the subgroup of non-inbred faculty get early tenure in a lesser proportion than men and inbred faculty. In order to check for a possible interaction between academic origin and gender, we also conducted two separate logistic regressions for the inbred and non-inbred subgroups (not shown here) introducing gender and the productivity measures as independent variables. We did not find such an interaction: male scientists, whether inbred or not, had more than double the odds of getting early tenure than females.[32]

6. Conclusions

This paper has explored academic careers in their early stages, especially early tenure, in the Spanish university and research system and their interaction with scientific production and mobility. Our analysis has addressed a key issue in the study of academic science: the relationship between performance and reward. We have approached this link considering tenure as a major form of reward and taking into account the effect of mobility on both performance and reward. In accordance with most of the literature, our paper showed support for the hypotheses of cumulative advantage in the sense that the single most important factor for explaining scientific output in the later stages of a career is early publication. Institutional research quality also seems to positively affect the academic performance of individuals who are granted tenure. Whether this is because selection committees in these universities apply more demanding criteria or because these institutions train and attract more productive researchers is an issue for further study. In a context of generally low mobility, a significant aspect of the analysis is that, however, inbred faculty had not performed worse than outsiders at the time of getting tenure. A hypothesis that would be worth exploring is whether this lack of differences in performance actually reflects certain environmental advantages of the inbred, in that their early non-mobile careers imply they have had less organizational transaction costs, and their permanence in the same department over time might have increased the likelihood of being involved in team work and of publishing co-authored papers within the group.

However, one interesting finding was related to the better performance of those who take part in stays at institutions abroad and return afterwards; this type of mobility is often sponsored by the department of origin and there are mutual expectations of return. The majority of Spanish universities are recruiting their own graduates but these do not get tenure with fewer publication merits than PhDs from different academic origins, indicating common socialization to norms emphasizing research and publication. Can we conclude then that seemingly universalistic rules are applied evenly in the tenure processes of Spanish universities and research institutions? Findings of the second part of our analysis preclude such a strong claim.

Embedded in an institutional context of low mobility and limited competition, internal job market dynamics operate and strongly affect the timing of academic rewards. Early tenure in Spain is explained by career trajectory and the social construction of the market more than by the previous performance measured by publication output, which has no positive effect in the probabilities of getting tenure shortly after being awarded a PhD. Our

27 This finding is contrary to McGee's argument (McGee, 1960) (based on very limited evidence and without multivariate analysis) and the Hargens and Farr (Hargens and Farr, 1973) findings that inbred scholars were discriminated against in terms of promotion and other rewards. In our sample, 3 years after the PhD, 35% of all inbred were tenured, whereas for non-inbred the percentage was only 18%.

28 The extent to which low mobility patterns are encouraged by departmental expectations is something for further data collection and analysis.

29 This is consistent with what is found for France (Gaughan and Robin, 2004).

30 Long et al. (Long et al., 1993) found a similar but less marked difference in their analysis of sex differences in the promotion from assistant to associate professors of a sample of bio-chemists. In our case, we did not have study data on family variables, or on differential access to qualified mentors and collaborators, all of which are important factors that the literature has shown to account for gender differences in career advancement of those who had already entered academia. In our sample, three years after being awarded a PhD, 33.6% of males had a permanent position, while the figure for women was only 16.3%.

31 We also ran a Logit model exclusively for engineers and technological scientists and we found that non-mobile careers had a significantly high positive impact on the odds of early tenure among engineers. The four mobility measures remained significant in engineering: engineering scientists who get tenure in their doctorate university, those who do not move in the year following the PhD, and those who do not have international post-doctoral experience or never left academia temporarily have greater odds of early tenure than those who moved. We have already mentioned the likely strong retention strategies of these departments that face fierce competition coming from the private sector.

32 We also ran separate logistic models by gender (not shown here) introducing the same independent variables as in the general model to analyze the likelihood of early tenure for men and women separately. The most important finding is that for female academics the inbreeding status variable is not significant. Another variable that is not significant in the case of female researchers is temporary mobility outside academia.

data show that, in an institutional context of very fragmented areas, post-doctoral mobility (either national or international, and especially mobility that implies leaving academia temporarily) does not pay. On the contrary, what speeds up the academic career in these early phases is permanence and institutional commitment. This is especially the case in the fields where supply does not exceed demand or in areas where the opportunity costs of private sector employment are high. In both cases, strong retention strategies based on job security take place from the departments. Despite the existence of gender-neutral institutional rules for evaluating performance and getting tenure at Spanish universities and research centers, our findings suggests the persistence of gender inequality: net of other variables, women advance more slowly towards tenure than their male counterparts.

We believe a number of research and policy implications can be drawn from this country case study. Although different forms of mobility are often claimed as a way of achieving research collaboration and other scientific gains, our findings for Spain question a long lasting assumption that mobility has a positive effect on career advancement. We have argued the need to analyze academic and scientific careers and labor market outcomes in the context of the institutional and structural rules governing the systems. The existence of open, competitive academic research job markets should not be taken for granted in the interpretation of the data regarding mobility and performance.

We had suggested the existence of diverse models of relationships between mobility, scientific performance and job placement, depending on the institutional arrangements. Our data provides us with clear insight into a system operating in Spain (and in many other countries) which is different to that characterized by open labor markets and mobility; a system that highly values institutional loyalty and rewards it with early lifelong job security, but one which, however, does not generally grant tenure against the criteria of scientific output.[33] The rigidities in the management of human resources and in the negotiation of individual conditions are overcome with an organizational strategy to increase the retention of new PhDs. Of course, the issue of the relative scientific performance of a model based on early retention strategies compared to others, in a similar context of resources, can only be addressed through systematic comparative studies.

The lack of international and inter-institutional mobility in some national contexts has been a policy issue for decades in the European context yet policy instruments have been focused on removing individual financial barriers (in the form of mobility fellowships) rather than on transforming the incentive structures of employing organizations. Policy makers still approach the issue of the creation of European research markets from the supply side of the researchers, and not from the side of organizational strategies and the institutional factors that affect those markets. An alternative approach would be to increase competition within and between universities and provide them with more flexible regulatory frameworks that allow for differential research faculty management. In order to improve the functioning of the external labor markets we first need a significant transformation of the organizational strategies and a reduction of the institutional rigidities; learning from history, it is our contention that only the differentiation and competition of autonomous institutions could consolidate practices that promote mobility and take advantage of mobility as a carrier of knowledge.

In this paper we have focused on careers until tenure. Further research is needed to analyze the relationship between mobility and performance at later stages of careers and to test whether the same patterns found here persist. If the relative environmental and social advantages of the inbred scientists are lost over time (progressively equalized with those of the outsiders), it would be plausible to find a curvilinear performance function. Likewise, we need to further understand the role of diverse types of collaboration (internal or external to one's institutional location) and their interactions with the different types of mobility. We have largely concentrated on how much (or how little) mobile and non-mobile researchers accomplish, but are the content and impact of what they do different? We believe these are important questions for future research.

[33] However, in contexts of strong departmental commitment with the internal candidates (regardless of their performance) a reputational "deterrent" effect could emerge that might preclude the most productive external candidates from applying.

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